

Toby Hurrell

Professor Cardona

MTEC-340-001

2 May, 2025

Post-Processing in Unity

Post-processing in Unity is an easy, versatile, and efficient way to modify or stylize the visual look of your Unity project. I will break down the topic into a couple of different sections; setup, color correction, camera effects, general effects, and processing algorithms.

Setup

Post-processing is available in almost any Unity project (besides Built-in Render Pipeline projects, which need the Post-Processing Version 2 package), but there are some settings you should be aware of before implementing any effects. Consider the render pipeline you would like to use with your project, as processing effects and quality differ depending on the pipeline. The Universal Render Pipeline (URP) allows for a range of graphical fidelities, but does not look as high fidelity as the High Definition Render Pipeline (HDRP). On the other hand, URP is considerably lighter on computation than HDRP, and thus runs on a larger variety of hardware. Both URP and HDRP allow for post-processing, but HDRP has more post-processing effects included than URP.

To start, toggle on Post-Processing on your Camera object. From there, you can either create a global or local (localized to an area based on a collider) volume. Right click or click create new in your hierarchy, hover on volumes, and select your desired object. In the volume info in the inspector, click “New” in the profiles section. This will create a post-processing profile which you can store visual effects and their parameters in. In your render pipeline

settings, you can choose whether or not to enable HDR, which will give you a wider range of color/lighting if enabled. To add an effect, click add override in your volume's inspector, and then add your desired effect.

Color Correction

Color correction effects include tonemapper, white balance, color adjustment, channel mixer, color curves, shadows midtones and highlights, lift gamma and gain, split toning, and the color lookup table. Color correction effects are used to modify how light and color are processed/perceived in your game. For instance, with color curves, you can map all colors to have a luminance of 0 besides red, which will make only red hues be perceived in your game. The color lookup table is the least intuitive—it allows for LUT files to be used, thus importing a particular color correction scheme onto your game from a premade file.

Camera Effects

Camera effects include depth of field, lens distortion, motion blur, and panini projection. These effects deal with the perception of space or the visual phenomenon of cameras. For instance, depth of field allows for a sense of camera focus to be added, while motion blur blurs the camera's image based on movement.

General Effects

These effects include film grain, bloom, vignette, chromatic aberration, and screen space lens flare. These can highly stylize the look of a game.

Processing Algorithms

In the Camera's inspector, both dithering and anti-aliasing can be turned on. Dithering will smooth visual quantization distortions, and anti-aliasing will smooth jagged edges by adding interpolation.

References

“EVERY Image Effect in Unity Explained - Post Processing v2 Tutorial” by Brackeys

<https://www.youtube.com/watch?v=9tjYz6Ab0oc>

Unity Documentation - Introduction to Post Processing

<https://docs.unity3d.com/Manual/PostProcessingOverview.html#effect-availability-and-location>

Unity Documentation - Post-Processing Effect Availability Reference

<https://docs.unity3d.com/Manual/post-processing-effect-availability-reference.html>